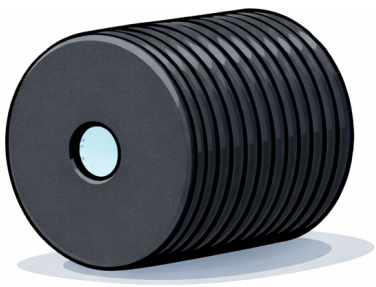
OpenMV® M8 Telephoto Lens

Datasheet

SKU: IR-M8-Telephoto-Lens / M8-Telephoto-Lens

Description

The M8 Telephoto Lens gives your OpenMV AE3 the ability to see longer distances. The standard lens has about a 70-degree FOV which allows your camera to see more things, but with less detail. This lens has about a 55-degree FOV, providing about a 1.4x zoom on any particular object. The lens comes with damping grease; gluing is recommended for vibration environments. Available with or without a 650 nm IR cut filter.



Target Areas

Long-range detection, eye tracking, license plate reading, quality inspection, wildlife monitoring, perimeter security.

Key Features

- **55-degree FOV (1.4x zoom)** — 54.2° diagonal / 47.0°H / 30.4°V with PAG7936. 1.43x optical zoom.
- **Ultra-low distortion** — Only -0.4% TV distortion for clean, undistorted images.
- **2 MP resolution** — Supports up to 2 MP sensor resolution.
- **M8 x P0.35 mount** — Standard M8 threaded mount for OpenMV AE3 and PAG7936 dual-camera modules.
- **IR cut filter option** — Available with 650 nm IR cut filter or without for night vision use.
- **3D model available** — STEP file available on GrabCAD for mechanical integration.

Applications

The M8 Telephoto Lens is great for any application where you need your OpenMV AE3 to zoom in on some subject. For example, the telephoto lens makes it easy to do eye-tracking by zooming in on someone's eye, making it easier to determine the direction their pupil is pointing.

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1 Specifications

1.1 Optical Specifications

Component	Details
Model	M8-4.42mm-S
Mount	M8 x P0.35 threaded
F-Number	F2.4
Effective Focal Length	4.42 mm
Back Focal Length	2.21 mm
Total Track Length	8.6 mm
Image Circle	Ø6.8 mm
Image Format	1/3"
Resolution	2 MP
Size	Ø8.0 x 8.0 mm
Construction	4E + IR (4 elements plus IR filter)
Focus / Iris / Zoom	Manual focus, fixed iris, fixed zoom
IR Coating Filter	650 nm (variant with IR cut filter)
Min. Object Distance	0.2 m

1.2 Field of View (1/3" format)

Component	Details
Diagonal	67°
Horizontal	60°
Vertical	36°
TV Distortion	-0.4%
Relative Illumination	>50%
Chief Ray Angle	<15.3°

All specifications ±5% tolerance.

1.3 Actual FOV with PAG7936 Sensor

Component	Details
Diagonal	54.2°
Horizontal	47.0°
Vertical	30.4°
Optical Zoom	1.43x

1.4 Compatibility

Component	Details
Supported Cameras	OpenMV AE3, dual-camera modules using PAG7936 sensors.
Installation	Includes damping grease. Gluing recommended for vibration environments. M8 lock ring solution in development.

2 General Info

Dimension	Value
Length	8 mm
Width	8 mm
Height	8 mm
Weight	1 g
Operating Temperature	-20°C to +65°C
Storage Temperature	-20°C to +65°C
Country of Origin	China
HS Code	8473.30.2000
UPC	608275911276, 608275911269
SKU	IR-M8-Telephoto-Lens, M8-Telephoto-Lens
ECCN	EAR99

3 Company Information

Field	Details
Website	https://openmv.io
Documentation	https://docs.openmv.io
Support	https://forums.openmv.io

4 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/m8-telephoto-lens
3D Model (STEP)	https://grabcad.com/library/m8-telephoto-lens-1

5 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.